

IP Addressing & Subnetting

Task (1):

IP Address Classes:

IP Class	IP Address Range	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Very large networks	A large ISP network may use Class A addressing.
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	A large university or company network may use Class B addressing.
Class C	192.0.0.0 – 223.255.255.255	Small networks	A home router/LAN may use a private Class C network such as 192.168.1.0/24.
Class D	224.0.0.0 – 239.255.255.255	Multicast communication	Video/audio streaming services can use multicast addresses such as 224.0.0.1.
Class E	240.0.0.0 – 255.255.255.255	Experimental/reserved use	Used for research and experimental networking , not normally assigned to everyday devices.

Task (2):

For the IP address **192.168.10.45** with subnet mask **255.255.255.0 (/24)**:

- **Step 1: Calculate the Network Address:**

Convert the last octet to binary:

- IP: **192.168.10.45**
- Last octet: **45 → 00101101**
- Subnet mask: **255.255.255.0**

- **Step 2: Calculate the Broadcast Address:**

Network: **192.168.10.00000000**

Broadcast: **192.168.10.11111111**

11111111 in decimal = **255**

✓ **Broadcast Address = 192.168.10.255**

- **Step 3: Calculate the Valid Host Range:**

First valid host = **192.168.10.1**

Last valid host = **192.168.10.254**

✓ **Valid Host Range = 192.168.10.1 – 192.168.10.254**

Final Answer:

Item	Answer
IP Address	192.168.10.45
Subnet Mask	255.255.255.0
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254
Number of Usable Hosts	254

Task (3):

❖ Subnetting Plan for a Food Delivery Startup:

- The startup has the network **10.0.0.0/24** and needs **4 separate subnets**. We can divide the **/24** network into four **/26** subnets.
- A **/26** subnet provides **64 IP addresses**, with **62 usable host addresses** per subnet.

Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Original network:

10.0.0.0/24

To create **4 subnets**, borrow 2 host bits:

/24 --} /26

No. of subnets:

$2^2 = 4$ subnets

Addresses per subnet:

$2^6 = 64$ addresses

Usable Hosts:

$64 - 2 = 62$ hosts

Final Subnet Structure

10.0.0.0/26	Delivery Partners
10.0.0.64/26	Restaurants
10.0.0.128/26	Customers
10.0.0.192/26	Admins

Task (4):

Step 1: Calculate the bits to borrow

- $2^5 = 32$ (enough) $\rightarrow$ borrow 5 bits

Step 2: New prefix length = $16 + 5 = /21$

Step 3: Convert /21 to a subnet mask

- /21 means 21 bits are network bits $\rightarrow$ third octet uses 5 of its 8 bits for subnetting:
 $11111000 = 248$
- Subnet mask = 255.255.248.0

Step 4: Calculate hosts per subnet:

- **Remaining host bits = $32 - 21 = 11$ bits**
- **Hosts per subnet = $2^{11} - 2 = 2048 - 2 = 2046$ usable hosts per subnet**

Task (5):

Why 192.168.1.300 won't connect:

- The error is that 300 is not a valid value for an IP address octet — each of the four numbers in an IPv4 address must be between 0 and 255 (since each octet is only 8 bits, max value $2^8 - 1 = 255$). Your friend's device likely has a typo or misconfigured static IP.
- A valid address in that same subnet (192.168.1.0/24) would be anything from 192.168.1.1 to 192.168.1.254 — for example, 192.168.1.30 or 192.168.1.100 would both work fine.